Exercise :
vector Spaces
Linear Combination
Example 1:-

$$V = \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix}$$

Determine above V is linear combinations of

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad V_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}, \quad V_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}$$

Not: If system is consistent then we can write Linear Combination otherwise we can't.

$\therefore \alpha V_1 + \beta V_2 + \gamma V_3 = V$

$$\alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} -3 \\ 3 \\ -2 \end{bmatrix} + \gamma \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix}$$

Take Augmented Matrix.

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & -1 \\ 0 & 3 & 7 & 1 \\ 1 & -2 & 5 & 10 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & -1 \\ 0 & 3 & 7 & 1 \\ 0 & 0 & 11 & 11 \end{array}\right] \quad R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & -1 \\ 0 & 1 & 7/3 & 1/3 \\ 0 & 0 & 11 & 11 \end{array}\right] \quad R_2/3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -6 & -1 \\ 0 & 1 & 7/3 & 1/3 \\ 0 & 0 & 1 & 1 \end{array}\right] \quad R_3/11$$

$x_1 - 2x_2 - 6x_3 = -1 \rightarrow (1)$

$x_2 + \frac{7}{3} x_3 = \frac{1}{3} \rightarrow (2)$

$\boxed{x_3 = 1} \rightarrow (3)$

put eq (3) in eq (2)

$$x_2 + \frac{7}{3}(1) = \frac{1}{3}$$

$$x_2 = \frac{1}{3} - \frac{7}{3}$$

$$\boxed{x_2 = -2}$$

put $x_2 = -2$ and $x_1 = 1$ in (1)

$$x_1 - 2(-2) - 6(1) = -1$$

$$x_1 + 4 - 6 = -1$$

$$x_1 - 2 = -1$$

$$\boxed{x_1 = 1}$$

Coz system is consistent and yield a unique solution so vector V is linear combination of V_1, V_2, V_3

$$V = V_1 - 2V_2 + 1V_3$$

Example : 2

$$V = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix}$$

is a LC of

$$V_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \quad V_2 = \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$$

$\alpha v_1 + \beta v_2 + \gamma v_3 = v$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -5 \\ -2 & 5 & 0 & 11 \\ 2 & 5 & 8 & -7 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & -5 \\ 0 & 5 & 4 & 1 \\ 0 & 5 & 4 & 3 \end{array}\right] \begin{array}{l} \\ R_2+2R_1 \\ R_3-2R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -5 \\ 0 & 1 & 4/5 & 1/5 \\ 0 & 5 & 4 & 3 \end{array}\right] \; R_2/5 \qquad 3-\cancel{5}\left(\frac{1}{\cancel{5}}\right)$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -5 \\ 0 & 1 & 4/5 & 1/5 \\ 0 & 0 & 0 & 2 \end{array}\right] \; R_3-5R_2$$

as $0 \neq 2$ so c inconsistent and we can write linear combination;

Example:3

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

is a LC of

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad M_2 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \qquad M_3 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

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$$xM_1 + yM_2 + zM_3 = A$$

$$x\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} + y\begin{bmatrix}0 & 1\\1 & 1\end{bmatrix} + z\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & 0\end{bmatrix}$$

$$\begin{bmatrix}x+z & y+z\\y+z & x+y+z\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & 0\end{bmatrix}$$

$x+z=1$

$y+z=1$

$y+z=1$

$x+y+z=0$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 1\\0 & 1 & 1 & 1\\1 & 1 & 1 & 0\end{array}\right]$$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 1\\0 & 1 & 1 & 1\\0 & 1 & 0 & -1\end{array}\right] R_3 - R_1$$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 1\\0 & 1 & 1 & 1\\0 & 0 & -1 & -2\end{array}\right] R_3 - R_2$$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 1\\0 & 1 & 1 & 1\\0 & 0 & 1 & 2\end{array}\right] -R_3$$

$x + z = 1 \rightarrow (1)$

$y + z = 1 \rightarrow (2)$

$z = 2$

put $z=1$ in eq (2)

$y + 2 = 1$

$y = -1$

put $z = 1$ in eq (1)

$x + 2 = 1$

$x = -1$

as system consistent & found unique value so A is a LC of M_1, M_2 & M_3

$A = -M_1 - M_2 + 2M_3$

Linearly Independence. (1st Method)

$v_1, v_2, v_3, \dots, v_3$

↳ or only independent when of them doe'st depend on other.

$a_1v_1, a_2v_2, a_3v_3 + \dots + a_nv_n = 0$

↳ only independent when all scalar becomes 0 otherwise dependent

Example: 1

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} \quad \text{and} \quad V_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix}$$

are linearly independent or linearly dependent

$$a_1V_1 + a_2V_2 + a_2V_3 = 0$$

$$a_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 + a_3 = 0$$
$$a_2 + a_3 = 0$$
$$a_1 + a_2 + a_3 = 0$$
$$2a_1 + 2a_2 + 3a_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 2 & 2 & 3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \end{array}\right] \begin{array}{l} \\ R_3 - R_1 \\ R_4 - 2R_1 \\ \\ \end{array}$$

$$\begin{array}{c} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array} \begin{array}{l} \\ R_3 - R_2 \\ R_4 - 2R_2 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} -R_3 \\ \\ \\ \\ \end{array}$$

Leading Var $= a_1, a_2, a_3$

$a_1 + a_3 = 0 \rightarrow (1)$

$a_2 + a_3 = 0 \rightarrow (2)$

$a_3 = 0$

put $a_3 = 0$ in (1)	put $a_3 = 0$ in (2)
$a_1 + 0 = 0$	$a_2 + 0 = 0$
$a_1 = 0$	$a_2 = 0$

Since all Scalars are 0 so V_1, V_2 and V_3 are linearly Independent.

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Example : 2

$V_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, V_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \quad V_3 = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$

$V_4 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

are Linearly Independent

$a_1 V_1 + a_2 V_2 + a_3 V_3 + a_4 V_3 = 0$

$$\left[\begin{array}{cccc|c} 1 & -1 & -2 & 2 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 2 & 2 & 1 & 1 & 0 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -2 & 2 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 4 & 5 & -3 & 0 \end{array}\right] \quad R_3 - 2R_1$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -2 & 2 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 0 & -7 & -7 & 0 \end{array}\right] \quad R_3 - 4R_2$$

$$\left[\begin{array}{cccc|c} 1 & -1 & -2 & 2 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right] \quad -\frac{R_3}{7}$$

Leading var $= a_1, a_2, a_3$ Let
Free var $- a_4$ $a_4 = t$

$a_1 - a_2 - 2a_3 + 2a_4 = 0 \rightarrow (1)$
$a_2 + 3a_3 + a_4 = 0 \rightarrow (2)$
$a_3 + a_4 = 0 \rightarrow (3)$

Use eq (3) put $a_3 = -t$ & $a_4 = t$ in (2)
$a_3 + a_4 = 0$ $a_2 = 3t - t$
$a_3 = -t$ $a_2 = 2t$

Now put in (1)
$a_1 - 2t + 2t + 2t = 0$
$a_1 = -2t$

Since t can be any number so its linearly independent rather they are Linearly dependent.

Example : 3

$M_1 = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$, $M_2 = \begin{bmatrix} -1 & 2 \\ 3 & 2 \end{bmatrix}$

$M_3 = \begin{bmatrix} 5 & -6 \\ -3 & -2 \end{bmatrix}$ are linearly indepent.

$$a_1M_1 + a_2M_2 + a_3M_3 = 0$$

$$a_1\begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} + a_2\begin{bmatrix} -1 & 2 \\ 3 & 2 \end{bmatrix} + a_3\begin{bmatrix} 5 & -6 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a_1 - a_2 + 5a_3 & 2a_2 - 6a_3 \\ 3a_1 + 3a_2 - 3a_3 & 2a_1 + 2a_2 - 2a_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$a_1 - a_2 + 5a_3 = 0$$
$$2a_2 - 6a_3 = 0$$
$$3a_1 + 3a_2 - 3a_3 = 0$$
$$2a_1 + 2a_2 - 2a_3 = 0$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 0 \\ 0 & 2 & -6 & 0 \\ 3 & 3 & -3 & 0 \\ 2 & 2 & -2 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 0 \\ 0 & 2 & -6 & 0 \\ 0 & 6 & -18 & 0 \\ 0 & 4 & -12 & 0 \end{array}\right] \begin{array}{l} \\ R_3 - 3R_1 \\ R_4 - 2R_1 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 6 & -18 & 0 \\ 0 & 4 & -12 & 0 \end{array}\right] \begin{array}{l} R_2/2 \\ \\ \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R_3-6R_2 \\ R_4-4R_2 \\ \\ \end{array}$$

Leading var $= a_1, a_2$ Let

Free var $= a_3$ $a_3 = t$

$a_1 - a_2 + 5a_3 = 0$ | $a_2 - 3a_3 = 0$

$a_1 - 3t + 5t = 0$ | $a_2 = 3t$

$a_1 = -2t$

Since $t \in \mathbb{R}$

and has infite many solution so

Lineary Dependent

Linear Independence & Determinant (2nd Method)

Example: 5

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} \right\}$$

are linearly independent.

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put vectors in column form

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

expand along C_1

$$\det(A) = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$\det(A) = (-1)^{1+1} 1 \begin{vmatrix} 2 & 4 \\ 4 & 5 \end{vmatrix} + 0 + (-1)^{3+1} 3 \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix}$$

$\det(A) = 1(10-16) + 3(4-2)$

$\det(A) = -6 + 6$

$\det(A) = 0$

As $\det(A) = 0$ so S is Linearly dependent set.

Exercise - Set

Question - 9 - 10

Determine whether given matrix are linearly independent

9 -

$$v_1 = \begin{bmatrix} 3 \\ -1 \\ -1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$\left[\begin{array}{ccc|c} 3 & 1 & 3 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 2 & 0 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 3 & 1 & 3 & 0 \\ -1 & 2 & 0 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \quad R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 3 & 1 & 3 & 0 \\ -1 & 2 & 0 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \quad -R_1$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \begin{array}{l} \\ R_2-3R_1 \\ R_3+R_1 \\ R_4-2R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right] \begin{array}{l} \\ R_3-2R_2 \\ R_4-R_1 \\ \\ \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R_4+R_3 \\ \\ \\ \end{array}$$

Leading var $= a_1, a_2, a_3$

$a_1 + a_3 = 0 \rightarrow (1)$

$a_2 = 0$

$a_3 = 0$

$a_1 + a_3 = 0$

$a_1 + 0 = a$

$a_1 = 0$

Since a_1, a_2 and a_3 all are 0 so V_1, V_2 and V_3 are Linearly independent

DATE:

10-

$$v_1 = \begin{bmatrix} -2 \\ -4 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 3 \\ -4 \\ 0 \\ 4 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -1 \\ -12 \\ 2 \\ 6 \end{bmatrix}$$

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0$$

$$\left[\begin{array}{ccc|c} -2 & 3 & -1 & 0 \\ -4 & -4 & -12 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 4 & 6 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -4 & -4 & -12 & 0 \\ -2 & 3 & -1 & 0 \\ 1 & 4 & 6 & 0 \end{array}\right] \quad R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & 3 & 3 & 0 \\ 0 & 4 & 4 & 0 \end{array}\right] \quad \begin{array}{l} R_2 + 4R_1 \\ R_3 + 2R_1 \\ R_4 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 3 & 3 & 0 \\ 0 & 4 & 4 & 0 \end{array}\right] \quad -R_2/4$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} R_3-3R_2 \\ R_4-4R_2 \end{array}$$

Leading Var = a_1, a_2 Let

Free var = a_3 $a_3 = t$

$a_2 + a_3 = 0$	$a_1 + 2a_3 = 0$
$a_2 + t = 0$	$a_1 + 2t = 0$
$a_2 = -t$	$a_1 = -2t$

$t \in \mathbb{R}$ has infinite many solution

So v_1, v_2, v_3 are Linearly Dependent.

Question : 19-20

Without solving a linear system why column vectors of matrix A are linearly dependent.

Question : 19

a-

$$A = \begin{bmatrix} -1 & 2 & 5 \\ 3 & -6 & 3 \\ 2 & -4 & 3 \end{bmatrix}$$

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If we see C_2 of A its obtained by

$$C_2 = -2C_1$$

So C_2 is depend on C_1 so we can say they are Linearly Dependent.

b-

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

If we see C_3 its obtained by

$C_3 = C_1 + C_2$

So vectors are Linearly Dependent.

Question: 20

a-

$$A = \begin{bmatrix} 4 & -1 & 2 & 6 \\ 5 & -2 & 0 & 2 \\ -2 & 4 & -3 & -2 \end{bmatrix}$$

→ There are 4 columns in $\mathbb{R}^3$

→ Any set of more than 3 vectors in $\mathbb{R}^3$ is Linearly Dependent

→ So there are only 4 column and 3 rows columns are Linearly Dependent

b-

$$A = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 3 & 1 \\ -1 & -1 & 0 \\ 3 & 5 & 2 \end{bmatrix}$$

If we see C_2 its obtained by

$$C_2 = C_1 + C_3$$

So, vectors are Linearly Dependent

Question : 21

Determine value of a such that vectors -

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2 \\ a \\ 4 \end{bmatrix}$$

are Linearly independent

When we put vectors in Matrix we get square matrix (3x3) so we can use Determinant Method

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & a \\ 1 & 1 & 4 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & -1 & 2 \\ 2 & 0 & a \\ 1 & 1 & 4 \end{vmatrix}$$

Expand along R_1

$$\det(A) = 1\begin{vmatrix} 0 & a \\ 1 & 4 \end{vmatrix} - (-1)\begin{vmatrix} 2 & a \\ 1 & 4 \end{vmatrix} + 2\begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix}$$

$$\det(A) = 1(0-a) + 1(8-a) + 2(2-0)$$

$$\det(A) = -a + 8 - a + 4$$

$$\det(A) = 12 - 2a$$

as we are given vectors are Linearly independent so $\det(A) \neq 0$

$$0 \neq 12 - 2a$$

$$2a \neq 12$$

$$\boxed{a \neq 6}$$

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Imp Point:

In Linear Independence:-

→ We can Determinant Method only when Matrix is square (same rows columns) otherwise we will use 1st method.